often used in multi-class classification.

(Read 3.13)

Information provided is greater for unlikely events.

"The Earth has gravity." This is so probable that it is unnecessary to say. This communicates no new information.

"The Earth ~~no longer has gravity~~ will not have gravity tomorrow" would be quite informative if true.

Less likely events should have more information.

Independent events' information should add.

self-information

$$I(x) = -\log P(x) \quad (1)$$

[Remember $X \sim P$ means X obeys the distribution defined by P.

If we use log base 2, $I(x)$ is measured in bits.

Ex: Flipping a coin. I flip the coin once.
I get $X = H \leftarrow$ heads.

$$I(x) = -\log_2 P(H) = -\log_2 \frac{1}{2} = 1 \text{ bit}$$

Ex: Flipping a coin (cont.)

Assuming the coin is unbiased and the flips are independent, the probability of flipping a head a second time is also $\frac{1}{2}$ and is uninfluenced by the first flip. This is an example where the information of the result of the first and second flip should add.

$$\underbrace{I_1(H)}_{\text{self-information from first flip}} + \underbrace{I_2(H)}_{\text{self-information from second flip}} = 2 \text{ bits} \quad (2)$$

Ex: Spin opposite Entangled Pair Electron

If we have two quantum entangled electrons. They are entangled such that they have opposite spin.

$P_1(up) = 0.5$, $P_1(down) = 0.5$

$I_1(up) = -\log_2 0.5 = 1$ bit

Given that we measured "up" on the 1st electron,

$P_2(up) = 0$, $P_2(down) = 1$

$I_2(down) = -\log_2 1 = 0$ bits

self-information is for a single outcome.

Shannon entropy is the expectation across all outcomes

$$H(x) = E_{x \sim p}[I(x)] = -E_{x \sim p}[\log P(x)] \quad (4)$$

$$= -\sum_{x \sim p} P(x) \log P(x) \quad (5)$$

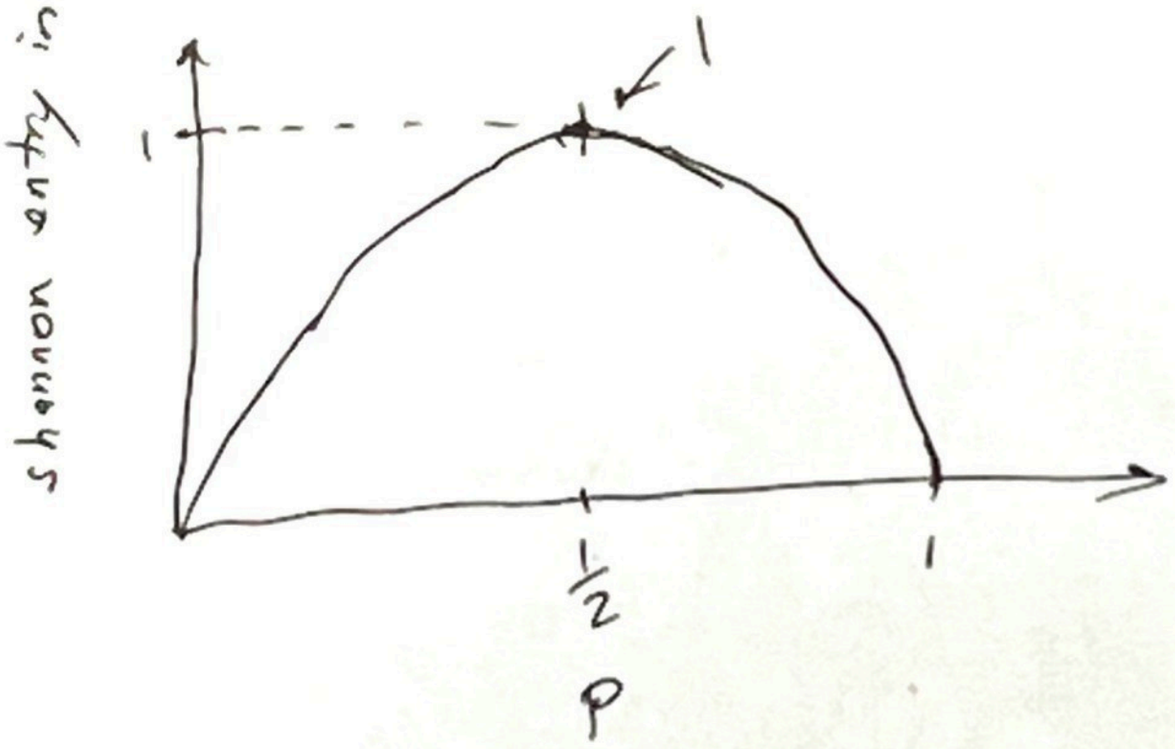
$H(x)$ = expected information from one sample drown from $x \sim P$.

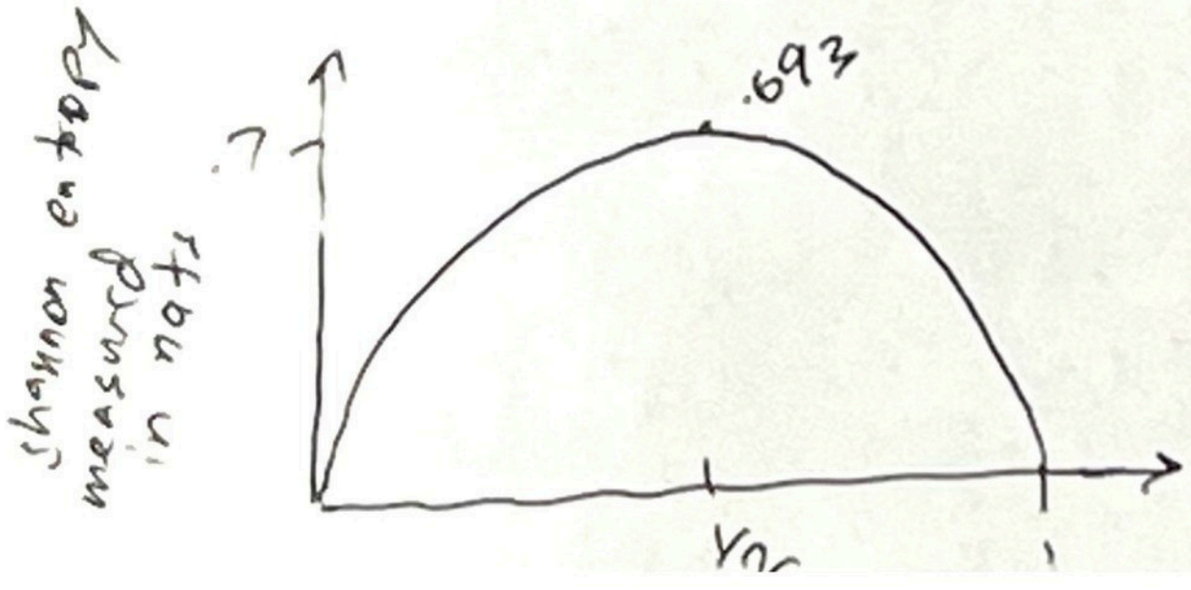
For the coin flip example

$$H(x) = -\sum_{x \in [H,T]} P(x) \log_2 P(x) = -P(H) \log_2 P(H) - P(T) \log_2 P(T)$$

$$= -\frac{1}{2} \log_2 \left(\frac{1}{2}\right) - \frac{1}{2} \log_2 \left(\frac{1}{2}\right) = -\frac{1}{2} \cdot (-1) - \frac{1}{2} \cdot (-1)$$

$$= \frac{1}{2} + \frac{1}{2} = 1 \text{ bit}$$





$$H(x) = -\sum_{x \sim p} P(x) \ln P(x)$$

in nats

natural log

CSCI6032 lecture 16 Info Theory and Cross Entropy Loss

Shannon Entropy (cont'd)

Example 2: The Less Known Shortcut

99% of the time ~~people~~ drivers take the main road

However there is a shortcut that few people know about which these few in the know take. Thus

1% of the time ~~people~~ drivers take the short cut.

we want to remember the exact pattern of people taking the main road vs. the short cut.

0 = main road

1 = shortcut

0 and 1 are symbols

0 0 0 0 1 0 0 0 1 0 1 1 0 0 0 0 0 0 0 0 0 1

Figure 1: strings of 0's and 1's

The result is characterized by occassional 1's appearing among long strings of zeroes and every once in a while we see more than one 1 in a row.

run of 9 ~~one~~ zeros

0 0 0 0 1 0 0 0 1 0 1 1 0 0 0 0 0 0 0 0 0 1

The self-information of 0 is

$$-\log_2 P(0) = -\log_2 0.99 = 0.014\ldots \text{ bits}$$

The self-information of 1 is

$$-\log_2 P(1) = -\log_2 0.01 = 6.64\ldots \text{ bits}$$

Shannon Entropy (cont.)

* The amount of information in a 1 is significantly higher than that of 0 simply because ones occur so infrequently. One could say that information is a measure of the amount of surprise.

If we encode the sequence of people taking respectively the main road vs. the shortcut as 0 and 1, then for 10000 ~~symbols~~ events we require 10000 symbols.

* Entropy is a measure of the expected information in a sequence of symbols.

More accurately stated, entropy is a measure of the expected amount of information per symbol in a sequence of symbols that follow distribution $P(x)$ where $x \in \mathcal{X}$ where $\mathcal{X}$ = the alphabet of allowed symbols.

↑ "script X"

$$H(x) = \underset{x \sim P}{E}[I(x)] = -\underset{x \sim P}{E}[\log P(x)]$$

$$= -\sum_{x \sim P} P(x) \log P(x) \quad (6)$$

For the less known shortcut, example (6) becomes

$$H(x) = -(P(0) \log P(0) + P(1) \log P(1)) \quad (7)$$

Substituting the values determined for self-information into (7) yields

$$H(x) = -(0.01 \overbrace{\log_2 0.01}^{6.64} + 0.99 \overbrace{\log_2 0.99}^{0.014})$$

Despite $6.64 >> 0.014$, it has far less impact than

$0.01 \log_2 0.01 = -0.066\ldots$

Thus,

$$H(x) = -(-0.066\ldots - 0.014\ldots) \approx 0.08 \text{ bits/symbol}$$

The expected ~~entr~~ information in 10000 drivers is ≈ 808 bits

However when we use 0 to denote taking the main road and 1 ~~for~~ the shortcut, it takes 10,000 bits. ~~to~~ Figure 1 clearly shows an inefficient encoding.

How might we encode driver choices in a way ~~get~~ closer to the 808 bits of entropy?

One possible encoding is run-length encoding. With run length encoding we represent the symbols as a sequence of runs.

AAAA BBBB B D FFFFFFFFFFFFF

becomes

4A5B1D11F

Shannon Entropy (cont.)

For the short cut example, we have runs of zeros and ones.

00000100010001

we could represent this as

(5,0), (1,1), (3,0), (1,1), (2,0), (1,1)

5 zeroes, one 1

← Figure 2

← Figure 3

However because ones are infrequent and runs of ones are even rarer, and because any symbol that is not a zero must be a one, we could choose to only encode the runs of zeros. This is called <u>zero run length encoding</u>.

Figure

With zero run length encoding ↑ 3 becomes

5, 3, 2

← Figure 4

The assumption is that after 5 zeros we have a one.

If we represent the 5, 3 and 2 using the same alphabet as our original sequence we get

101, 11, 10 ← We use comma as a special character

101, 011, 010 ← If we use fixed length fields each of 3 bits to denote run length.

If we use fixed length fields, we don't need the commas as delimiters, and this becomes

101011010

1st run, 2nd run, 3rd run

Figure 5

Shannon Entropy (cont.)

By using fixed length fields, we avoid the need for a special delimiter, but then what happens when we have a run of zeros so long that the field overflows, what do we do?

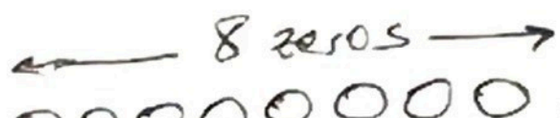


Figure 6

If we have a 3-bit field for denoting run length

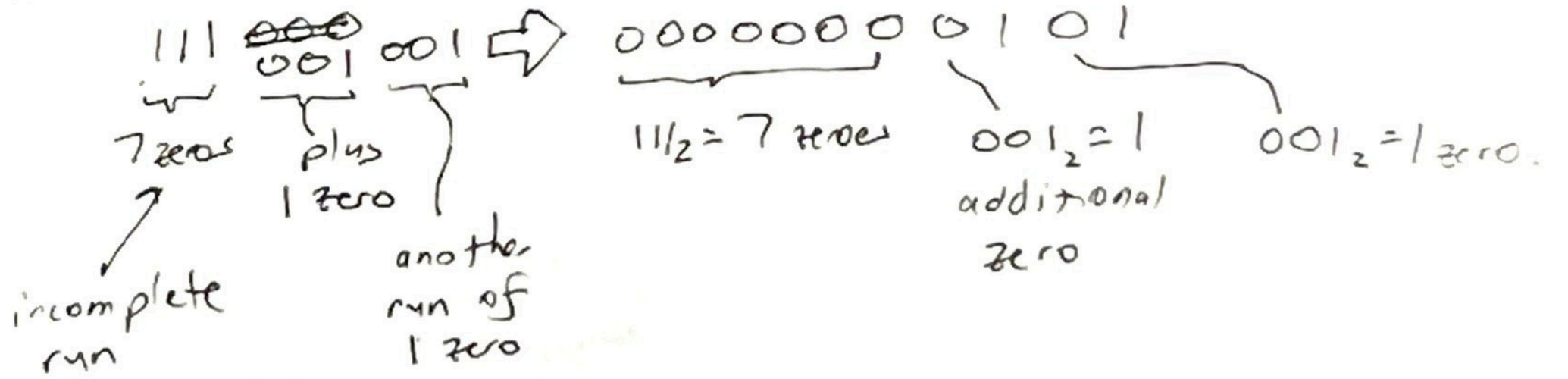
$111_2 = 7_{10}$ ← base 10

↑ base 2

Figure 7

$111_2 + 1 = 000_2$ ← overflow!

One common way to handle this is to ~~deal~~ treat 111_2 as a special code denoting an incomplete run. An incomplete run does not end in a one.



For this encoding w/ P(0) = .99 and P(1) = .01 we should get be able to represent such long runs of zeros efficiently. Let k = 6 denote the

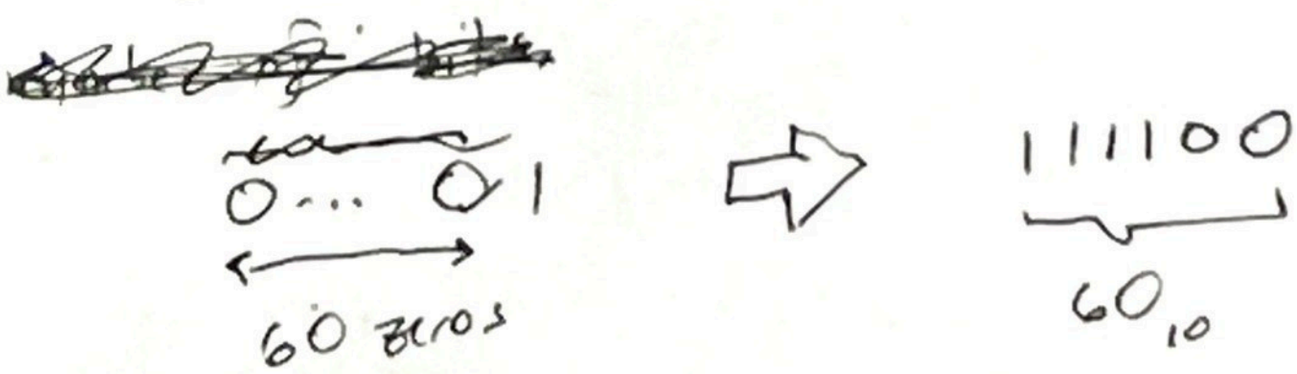


Figure 8

Shannon Entropy (cont.)

We reduced the number of bits representing 61 drivers down to 6.

compression ratio = $\frac{61}{6} \approx 10.2$

The number of bits per symbol if this holds out across a large file would correspond to .098 bits output per input symbol.

shannon entropy $H(x)$ was estimated $\approx$ 0.08 bits.

~~actual eff~~

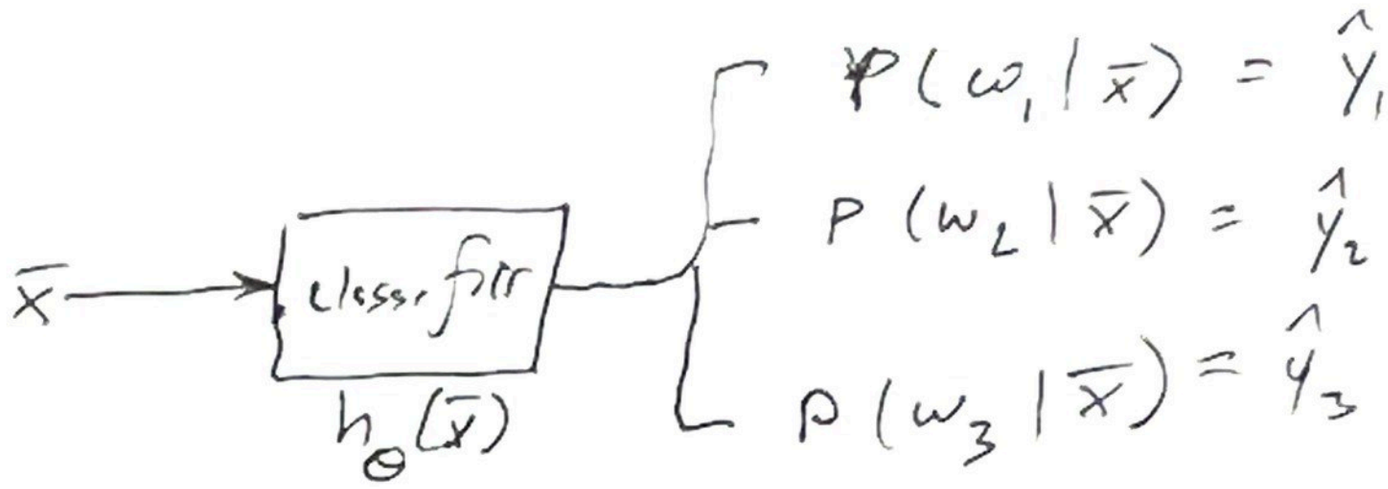
~~Finding the shannon entropy~~

shannon entropy is the optimal bound.

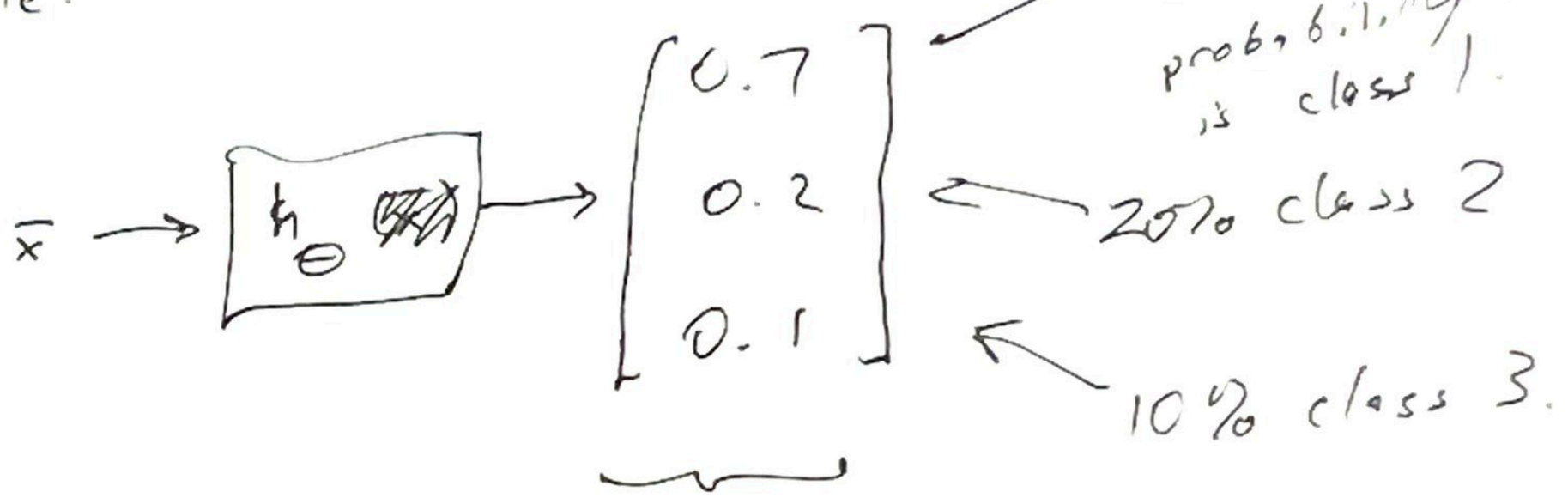
Finding the actual efficiency of zero run length encoding is left up to a homework exercise.

10 M

Multi-class classification



Example:



estimate of the posterior probabilities

estimated posterior probability of each class

Based on the predicted probability we make a decision.

If we are minimizing error then we pick the class with the maximum predicted probability.

CSCI6032 lecture 16 Info Theory and Cross Entropy Loss

Cross-entropy Loss

$$H(P,Q) = -\mathop{E}_{x\sim P} \log Q(x)$$

Instead of P and Q lets adopt notation from our lectures on classification, except we denote the K classes ~~ω_1, ω_2~~ (ω_1 is ω_K ~~ω_2~~)

$$P(\omega_k|\bar{x}) = \hat{y}_k$$

Let y = the true label as an integer

ω_1 corresponds to $y=1$, ω_2 corresponds to $y=2$

$$\mathcal{L}(y,\hat{y}) = H(y,\hat{y}) = -\sum_{k=1}^{K} \underbrace{\mathbb{I}(y=k)}_{\text{indicator function}} \log(\hat{y}_k)$$

Treating cross-entropy as a loss

aside

- y is a "one-hot" vector. When we know the truth that $y=3$, the probability that $y=3$ is 1. The probability that y is anything but 3 is zero.

$$P(y=k|y=i) = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{corresponds to } i=3$$

$\mathbb{I}$ is an indicator function. It is 1 only when $y=k$ and zero otherwise.

$$\mathbb{I}(y=k) = \begin{cases} 1 & \text{if } y=k \\ 0 & \text{if } y\neq k \end{cases}$$

(very similar to a Kronecker delta)

Example let $K = 3$ means there are 3 classes.

$\hat{y}_1 = 0.7$ (~~Classifier~~ Posterior probability is 70% for class 1)

$\hat{y}_2 = 0.2$ (Posterior probability is 20% for class 2)

$\hat{y}_3 = 0.1$ (" " 10% for class 3)

Let Assume $y = 2$ meaning the true label is 2.

$$\mathcal{L}(y=2, \hat{y}) = -[\mathbb{1}(y=1)^{0} \log(\hat{y}_1) + \mathbb{1}(y=2)^{1} \log(\hat{y}_2) + \mathbb{1}(y=3)^{0} \log(\hat{y}_3)]$$

$$\mathcal{L}(y=2, \hat{y}) = -\log(\hat{y}_2)$$

$$\mathcal{L}(y=2, \hat{y}) = -\log 0.2 \approx 1.609 \text{ nats}$$

Example let $K = 3$

$\hat{y}_1 = 0$

$\hat{y}_2 = 1$

$\hat{y}_3 = 0$

$$\mathcal{L}(y=2, \hat{y}) = -[\mathbb{1}(y=1)^{0} + \mathbb{1}(y=2)^{1} \log(\hat{y}_2) + \mathbb{1}(y=3)^{0} \log \hat{y}_3$$

$$= -\log(\hat{y}_2) = -\log 1 = 0$$

No nats of loss because classifier is confident $\bar{x}$ is of class 2 when it is of class 2.

With cross entropy loss, we experience some loss even when the class with the highest predicted probability is the correct class. When the classifier predicts correctly but is timid, we give it a little push confirming it is right

CORRECT

$$\hat{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\mathcal{L}(y=2, \hat{y}) = 0$

uncertain classifier

$$\hat{y} = \begin{bmatrix} 0.3 \\ 0.4 \\ 0.3 \end{bmatrix}$$

$\mathcal{L}(y=2, \hat{y}) =$

$-\log 0.4 \approx 0.9$

WRONG!

$$\hat{y} = \begin{bmatrix} 0.8 \\ 0.1 \\ 0.1 \end{bmatrix}$$

$\mathcal{L}(y=2, \hat{y}) =$

$-\log 0.1 \approx 2.3$

Cross-entropy vs. KL divergence

$H(P,Q)$ = the expected number of bits to encode message $\sim P$ with an optimal encoder for Q.

$H(P,Q) - H(P)$ = the expected extra bits needed to encode messages $X \sim P$ using an ~~the~~ optimal encoder of Q.

The expected extra bits is called the Kullback-Leibler (KL) Divergence (see 3.13 in Goodfellow et al)

$$\text{KL Divergence} = D_{KL}(P \| Q) \triangleq H(P,Q) - H(P) \quad (1)$$

In the Goodfellow book this is defined as

$$D_{KL}(P \| Q) = E_{x \sim p}\left[\log \frac{P(x)}{Q(x)}\right] \quad (2)$$

But (2) and (1) are equivalent. From (1)

$$H(P,Q) - H(P) = \left(-\underset{x \sim P}{E}[\log Q(x)]\right) - \left(-E_{x \sim P}[\log P(x)]\right)$$

$$= E_{x \sim p}[\log P(x)] - E_{x \sim p}[\log Q(x)]$$

$$= E_{x \sim p}\left[\log \frac{P(x)}{Q(x)}\right]$$

Q.E.D.